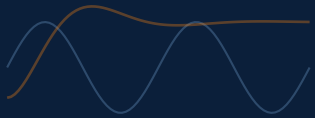
DAY 1 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Electrical Machines and Actuators

Principles, machine structures, motor models, transmissions and sizing + introduction to dynamic system modelling

Mechatronics Training Programme

Theoretical Course Series ▪ Day 1 of 5

Agenda and learning objectives

Code	Session	Time
1.1	Fundamental principles and classification	0.50 h
1.2	Structure of electrical machines	1.00 h
1.3	Electrical and mechanical models	1.00 h
1.4	Actuators and motion transmission	0.75 h
1.5	Characteristics and selection criteria	0.75 h
2.1	Modelling of dynamic systems (part 1)	1.00 h
	Net teaching time	**5.00 h**

Breaks (≈10 min) roughly every hour are additional.

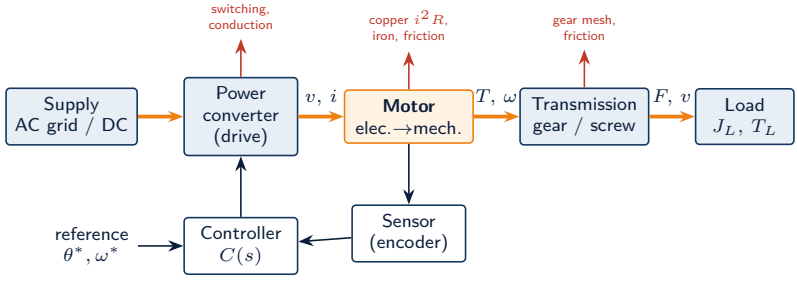
After today you can . . .

- explain how force/torque and voltage arise from $\boldsymbol{B}$, i and motion
- distinguish DC, BLDC, PMSM, induction and stepper machines
- write and use the DC-motor model, block diagram and transfer function
- reflect loads through gears and screws; choose a gear ratio
- size a servo motor from a move profile (peak, RMS, thermal)
- build lumped models of mechanical, electrical and coupled systems

Session 1

1.1 Fundamental principles and classification

Electromechanical energy conversion in a servo axis

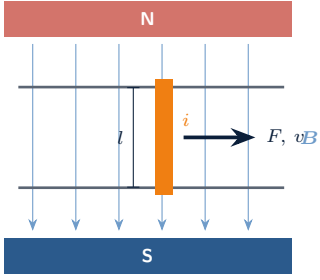


$$P_{\text{el}} = v\,i \xrightarrow{\text{motor}} P_{\text{mech}} = T\,\omega = P_{\text{el}} - P_{\text{loss}}$$

$$\eta = \frac{T\omega}{v\,i}, \quad \text{typical servo motor } \eta \approx 85\text{–}95\,\%$$

- Conversion is **reversible**: motoring and generating (braking).
- Coupling medium: the **magnetic field** in the air gap.
- Losses become heat ⇒ the **thermal limit** sets continuous torque.

Lorentz force and Faraday's law: the two halves of every motor



(B into the plane in the real geometry)

Lorentz force (conductor of length l): $\boldsymbol{F} = i\,\boldsymbol{l} \times \boldsymbol{B} \Rightarrow F = B\,l\,i$

Faraday / motional EMF: $e = -\dfrac{d\lambda}{dt} \Rightarrow e = B\,l\,v$

Power balance (ideal, lossless): $e\,i = (Blv)\,i = (Bli)\,v = F\,v$

For a coil with N turns at radius r ($2N$ active sides):

$$T = \underbrace{2NBlr}_{k_t}\, i, \qquad e = \underbrace{2NBlr}_{k_e}\, \omega$$

Consequence

The same geometry gives **torque per ampere** and **volts per rad/s**: in SI units $k_t = k_e$ (N m/A $\equiv$ V s/rad).

Reluctance force and the co-energy method



Iron is attracted towards the position of **minimum reluctance** (maximum inductance), even without magnets.

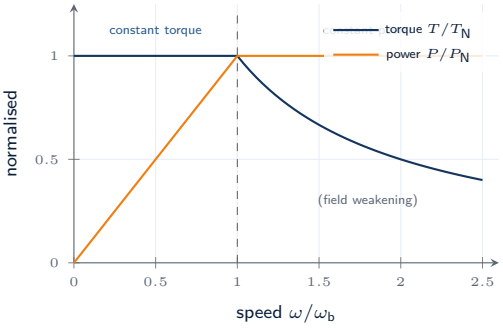
Co-energy: $W'(i,\theta) = \int_0^i \lambda(i',\theta)\, di'$, $\quad T = \left.\dfrac{\partial W'}{\partial \theta}\right|_{i=\text{const}}$

Linear magnetics, $\lambda = L(\theta)\, i$: $\quad T = \frac{1}{2}\, i^2\, \dfrac{dL}{d\theta}$, $\quad F = \frac{1}{2}\, i^2\, \dfrac{dL}{dx}$

- Torque $\propto i^2$: independent of current sign $\Rightarrow$ solenoids, switched-reluctance and stepper machines.
- $L(\theta) = L_0 + L_2 \cos 2\theta$ gives $T = -L_2\, i^2 \sin 2\theta$ (plot).
- In IPM synchronous motors a **reluctance torque** adds to the magnet torque (saliency $L_q > L_d$).
- With saturation, use the full co-energy W', not $\frac{1}{2} L i^2$.

[Fitzgerald, Kingsley & Umans, 2003, ch. 3]

Torque, speed and power



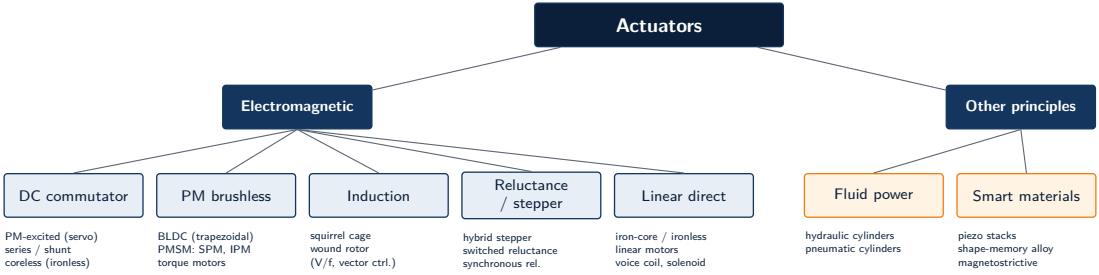
$$P = T\,\omega, \qquad \omega = \frac{2\pi n}{60} \quad (n \text{ in rpm})$$

$$P\,[\text{kW}] = \frac{T\,[\text{N m}] \cdot n\,[\text{rpm}]}{9549}$$

- Torque is limited by **current** (thermal, magnet demagnetisation).
- Speed is limited by **voltage** (back-EMF $k_e\omega$ must stay below the supply) and by mechanics.
- Above base speed ω_b, field weakening trades torque for speed at $\approx$ constant power.
- Linear motion: $P = F\,v$; screw/belt convert $T, \omega \leftrightarrow F, v$.

Example: 1.27 N m at 3000 rpm $\Rightarrow \omega = 314$ rad/s, $P = 400$ W.

Classification of motors and actuators



	PM servo motor	Hydraulic	Pneumatic	Piezo
Force/torque density	high	**very high**	medium	high (tiny stroke)
Bandwidth / precision	**high / high**	medium / medium	low / low	**very high** / nm
Controllability, cleanliness	**excellent**	leakage, pump	compressible air	HV drive

Session 1.1 – Summary / key take-aways

- A motor converts $v\,i$ into $T\,\omega$ through the air-gap magnetic field; the process is reversible.
- **Lorentz**: $F = Bli \;\Leftrightarrow$ **Faraday**: $e = Blv \Rightarrow k_t = k_e$ (SI).
- **Reluctance** torque $T = \frac{1}{2} i^2\, dL/d\theta$ needs no magnets (steppers, SRM, IPM saliency).
- Current limits torque; voltage limits speed; heat limits the continuous operating point.

Remember

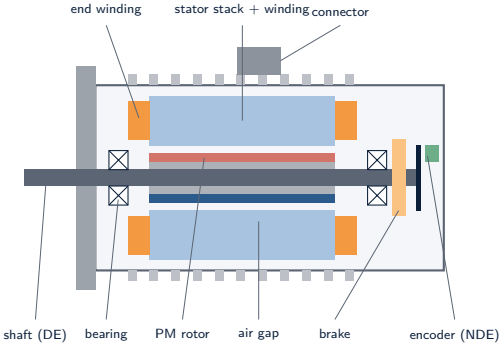
$$P = T\omega = Fv, \qquad \omega = \frac{2\pi n}{60}$$

$$T = k_t\, i, \qquad e = k_e\, \omega$$

Session 2

1.2 Structure of electrical machines

Anatomy of an industrial servo motor

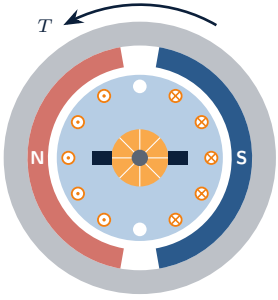


Schematic longitudinal section of a PM servo motor (not to scale).

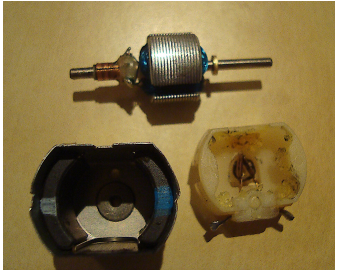
- **Stator**: laminated iron stack with slots and (three-phase) copper windings – the heat source is on the outside, close to the housing.
- **Rotor**: shaft + magnets (PM) or cage (induction); low inertia is a design goal for servo motors (long, slim rotors).
- **Air gap**: $\approx 0.3\,\text{mm}$ to $1\,\text{mm}$; all torque is produced here.
- **Feedback device**: encoder or resolver on the non-drive end (Day 5).
- **Holding brake** (optional, spring-applied), **bearings**, **shaft seal**, **connectors**.
- Ratings defined per IEC 60034-1 (duty types S1–S10), protection class IP65/67 typical.

Industrial "AC servo" usually means a **PMSM** with sinusoidal back-EMF, encoder and a matched drive.

Brushed (commutator) DC motor: construction



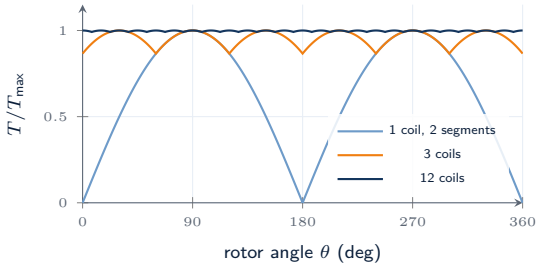
stator magnets, armature, commutator, brushes



Jfer91, CC BY-SA 3.0, via Wikimedia Commons

- Field from **stator magnets** (or field winding).
- Current-carrying **armature** rotates.
- **Commutator + brushes** reverse coil currents so that conductors under N always carry the same direction.
- Armature MMF stays **perpendicular** to the field $\Rightarrow$ maximum torque per ampere, automatically.

Mechanical commutation and torque ripple



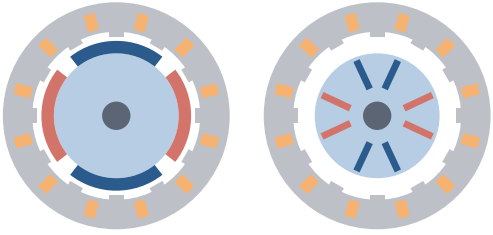
Idealised: only the best-aligned coil group is counted.

- The commutator is a **mechanical inverter**: it switches each coil's current as it passes the neutral zone.
- With n coil groups, peak-to-peak ripple $\approx 1 - \cos(\pi/2n)$:
  $n = 3$: 13 %, $n = 12$: 0.9 %.
- Commutation sparks, brush wear (typically a few thousand hours), EMI, speed limit from brush bounce.
- Heat is generated in the **rotor** $\Rightarrow$ poor cooling path, low continuous torque per volume.
- Simple drive: one H-bridge.

Idea for BLDC/PMSM

Replace the commutator by **power electronics + rotor-position sensor** $\Rightarrow$ electronic commutation.

Permanent-magnet brushless machines: BLDC and PMSM

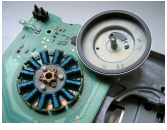


SPM IPM

SPM: magnets glued on surface, $L_d \approx L_q$

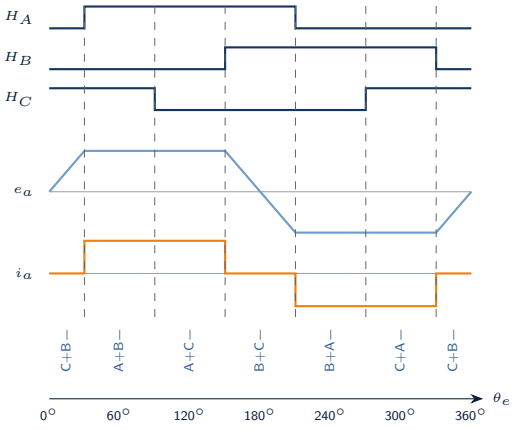
IPM: buried magnets, $L_q > L_d$, reluctance torque

- "Inside-out" DC motor: magnets on the **rotor**, three-phase winding in the **stator** $\Rightarrow$ heat close to the housing, no brushes.
- **BLDC**: concentrated windings, **trapezoidal** back-EMF, six-step (block) commutation.
- **PMSM**: distributed (or optimised) windings, **sinusoidal** back-EMF, sinusoidal / field-oriented control.
- p pole pairs: $\theta_e = p\,\theta_m, \quad \omega_e = p\,\omega_m.$
- Servo: mostly SPM; traction/spindles: IPM.



Sebastian Koppehel, CC BY 3.0, via Wikimedia Commons

Electronic commutation I: six-step (block) with Hall sensors

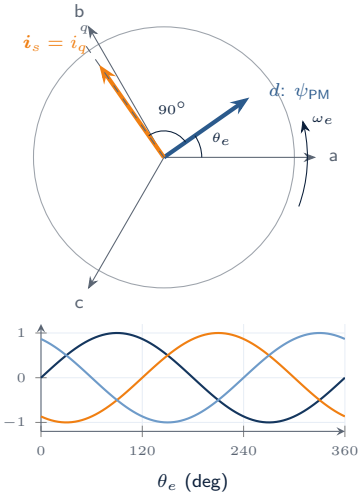


θ_e	$H_A H_B H_C$	on	current path
30–90	101	A+ B−	$i_a = -i_b = I$
90–150	100	A+ C−	$i_a = -i_c = I$
150–210	110	B+ C−	$i_b = -i_c = I$
210–270	010	B+ A−	$i_b = -i_a = I$
270–330	011	C+ A−	$i_c = -i_a = I$
330–30	001	C+ B−	$i_c = -i_b = I$

- Three Hall sensors give **60°_e resolution**: 6 states per electrical period.
- Two phases conduct, one floats:
  $$T = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m} = \frac{2EI}{\omega_m}.$$
- Current transfers at each step ⇒ **commutation torque ripple** (10–20 %).
- Sensorless option: detect back-EMF zero crossing in the floating phase.

Hall mounting/coding differs between manufacturers – table is for the alignment shown.

Electronic commutation II: sinusoidal commutation (PMSM)

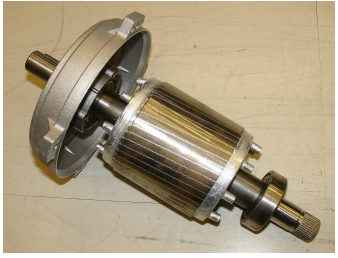


Impose phase currents **in phase with the sinusoidal back-EMF**:

$$e_x = k_{e,\text{ph}}\,\omega_m \sin(\theta_e - \varphi_x), \quad i_x = \hat{I}\sin(\theta_e - \varphi_x)$$
$$\varphi_x \in \{0, \tfrac{2\pi}{3}, \tfrac{4\pi}{3}\} \text{ for } x = a, b, c$$
$$T = \frac{\sum_x e_x i_x}{\omega_m} = k_{e,\text{ph}}\hat{I}\sum_x \sin^2(\theta_e - \varphi_x) = \tfrac{3}{2}\,k_{e,\text{ph}}\,\hat{I} = \text{const.}$$

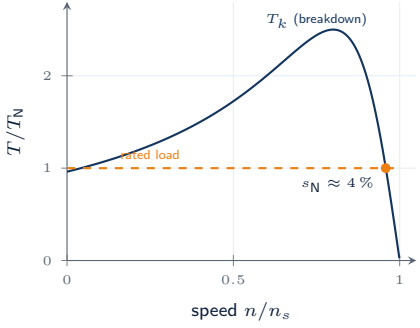
- Needs **continuous rotor angle** (encoder/resolver), not just 6 Hall states.
- Current vector kept at $90°_e$ to the magnet flux: $i_d = 0$, $T = \frac{3}{2}\,p\,\psi_{\text{PM}}\,i_q$.
- Ideally **ripple-free** torque ($< 2\,\%$ vs. 10–20 % for six-step).
- Implemented via Clarke/Park transforms = **field-oriented control** (Day 3–4).

Induction (asynchronous) motor: short introduction



Zureks, CC BY-SA 3.0, via Wikimedia Commons

- Rotating stator field induces rotor currents in a **squirrel cage**.
- Rotor must **slip** behind the field to see induced voltage.



$$n_s = \frac{60\,f}{p}, \qquad s = \frac{n_s - n}{n_s}$$

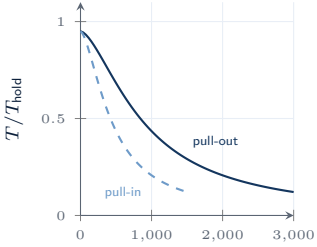
Kloss: $$\frac{T}{T_k} \approx \frac{2}{s/s_k + s_k/s}$$

- Robust, cheap, no magnets, field weakening easy.
- Rotor losses $\propto s$ heat the rotor.
- Servo use: spindles, large axes, with **vector control** + encoder.
- Plot: $s_k = 0.2$, $T_k = 2.5\,T_N$.

Stepper motor: short introduction



tony_duell, CC BY 2.0, via Wikimedia Commons



qualitative shape

- Hybrid stepper: toothed rotor with axial magnet, 2 phases.
- Step angle: $\theta_{\text{step}} = \dfrac{360^\circ}{2\,m\,N_r}$; $m = 2$, $N_r = 50 \Rightarrow$ **1.8°** (200 steps/rev).
- **Open-loop** positioning by counting steps; micro-stepping for smoothness.
- Torque falls with speed (L, back-EMF).
- Risk: **lost steps** if load torque exceeds pull-out torque; resonance at low/medium speed.

Comparison of machine types for motion control

	Brushed DC	BLDC	PMSM (AC servo)	Induction	Hybrid stepper
Commutation	mechanical	6-step, Hall	sinusoidal / FOC	slip, vector ctrl.	phase sequencing
Torque density	medium	high	**very high**	medium	high at low speed
Torque ripple	low–medium	medium–high	**very low**	low	high (detent)
Speed range	limited (brushes)	high	high	**very high**	low
Feedback	optional	Hall	encoder/resolver	encoder (servo)	none needed
Maintenance	brushes	none	none	none	none
Drive complexity	**H-bridge**	3-ph. inverter	3-ph. inverter + FOC	3-ph. inverter	2×H-bridge
Use	toys, small tools	fans, pumps, drones	robots, machine tools	spindles, pumps	printers, 3D printers

Rule of thumb for industrial servo axes: **PMSM** below ≈10 kW and where dynamics/precision dominate; induction for large spindles and rugged, high-speed applications.

Session 1.2 – Summary / key take-aways

- All machines: stator, rotor, air gap; torque comes from interaction of two magnetic fields (or reluctance).
- **Commutation** keeps stator and rotor fields $\approx 90^\circ$ apart: commutator (DC), six-step + Hall (BLDC), continuous angle + sinusoidal currents (PMSM).
- BLDC/PMSM: windings on the stator $\Rightarrow$ better cooling, no wear.
- Induction: needs slip; stepper: open-loop steps, risk of step loss.

Remember

$$\theta_e = p\,\theta_m$$

$$T_{\text{PMSM}} = \tfrac{3}{2}\,p\,\psi_{\text{PM}}\,i_q$$

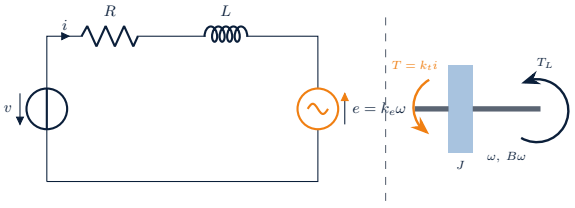
$$n_s = 60f/p, \quad s = (n_s - n)/n_s$$

$$\theta_{\text{step}} = 360^\circ/(2mN_r)$$

Session 3

1.3 Electrical and mechanical models

Equivalent circuit of the PM DC motor



Sym.	Meaning	Unit
R	winding (terminal) resistance	Ω
L	winding inductance	H
k_e	back-EMF constant	V s/rad
k_t	torque constant	N m/A
J	rotor (+ load) inertia	kg m^2
B	viscous friction coeff.	N m s/rad
T_L	load torque	N m

Sign convention: motor (consumer) convention – v and i positive into the terminals; ω and T positive in the same direction; T_L opposes motion. Valid for brushed DC and, per q-axis, for PMSM under FOC (Day 4).

Fundamental motor equations and why $k_t = k_e$

KVL: $v = R\,i + L\dfrac{di}{dt} + k_e\,\omega$

Newton: $J\dfrac{d\omega}{dt} = k_t\,i - B\,\omega - T_L$

Kinematics: $\dfrac{d\theta}{dt} = \omega$

Multiply KVL by i and Newton by ω:

$$\underbrace{v\,i}_{P_{\text{el}}} = \underbrace{R i^2}_{\text{Cu loss}} + \underbrace{\tfrac{d}{dt}\left(\tfrac{1}{2}L i^2\right)}_{\text{magn. energy}} + \underbrace{k_e\,\omega\,i}_{P_{\text{conv}}}$$

$$\underbrace{k_t\,i\,\omega}_{P_{\text{conv}}} = \underbrace{\tfrac{d}{dt}\left(\tfrac{1}{2}J\omega^2\right)}_{\text{kin. energy}} + \underbrace{B\omega^2}_{\text{friction}} + \underbrace{T_L\,\omega}_{\text{to load}}$$

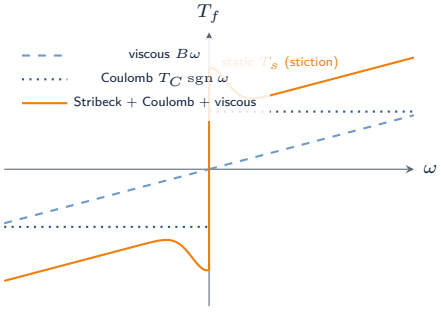
Energy conservation $\Rightarrow$ k_t=k_e

The converted power must be identical on both sides: $e\,i = T\,\omega$

$\Rightarrow\ k_e\,\omega\,i = k_t\,i\,\omega\ \Rightarrow\ \boldsymbol{k_t = k_e}$ (SI units).

- Datasheets often give k_e in V/krpm: $k_e[\text{V s/rad}] = k_e[\text{V/krpm}]\,/\,104.7$.
- Example: 10.47 V/krpm $\Rightarrow$ 0.1 V s/rad $\Rightarrow k_t = 0.1$ N m/A.
- Two linear, coupled 1st-order ODEs $\Rightarrow$ a **2nd-order** system from v to ω.
- k_t falls slightly at high current (saturation) and with magnet temperature ($\approx -0.1\,\%/\text{K}$ for NdFeB).

Friction and load torque

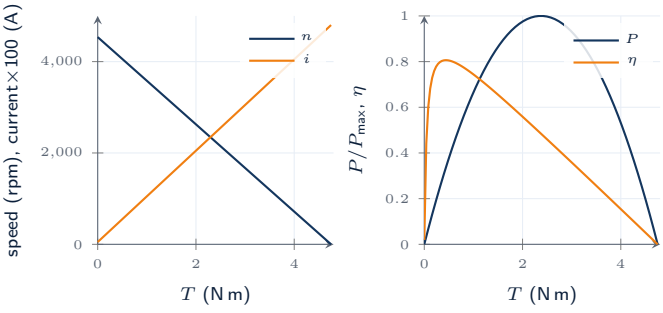


$$T_f(\omega) = \left[T_C + (T_s - T_C)\, e^{-(\omega/\omega_s)^2}\right] \operatorname{sgn}(\omega) + B\,\omega$$

- **Viscous** $B\omega$: linear, enters the transfer function.
- **Coulomb** T_C and **stiction** T_s: **nonlinear** – cause stick–slip, limit cycles and position errors at reversal (Day 2 linearisation, Day 4 compensation).
- **Load torque** T_L: process force, gravity (mgr on vertical axes – constant!), spring forces, cutting forces.
- In the linear model: T_L is a **disturbance input**, B lumps all speed-proportional losses.

[Armstrong-Hélouvry, Dupont & Canudas de Wit, 1994]

Steady-state characteristics at constant voltage



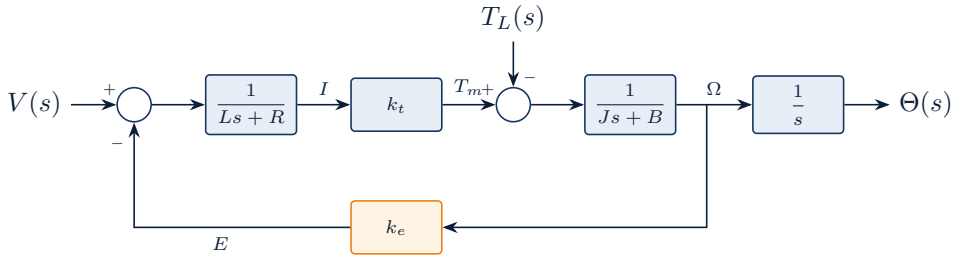
Example motor: $v = 48\,\text{V}$, $R = 1\,\Omega$, $k_t = k_e = 0.1$ SI, no-load friction $T_0 = 0.05\,\text{N m}$.

Set $d/dt = 0$:

$$\omega = \frac{v}{k_e} - \frac{R}{k_e k_t}\,T$$

- No-load speed $\omega_0 = v/k_e$ (4584 rpm ideal).
- Stall torque $T_{\text{st}} = k_t v/R$ (4.8 N m, 48 A!).
- Slope $-R/(k_e k_t)$: speed **droop** under load.
- $P_{\max} = v^2/(4R)$ at $T_{\text{st}}/2$ – never a servo operating point (50 % efficiency).
- $\eta_{\max} = \left(1 - \sqrt{i_0/i_{\text{st}}}\right)^2 \approx 81\,\%$ near light load.

Block diagram of the DC motor



- Two cascaded first-order lags: **electrical** (L, R) and **mechanical** (J, B).
- The back-EMF k_e is an **internal feedback loop** – it gives the motor its natural speed stabilisation (damping).
- T_L enters as a **disturbance** at the torque summing point.

Servo drives close a fast **current loop** around $1/(Ls+R)$, so that $T_m \approx k_t\, i^*$ – the motor then appears as a pure torque source acting on $1/(Js+B)$ (Day 4: cascaded current/speed/position loops).

Transfer functions and time constants

From the block diagram ($T_L = 0$):

$$G_{\omega v}(s) = \frac{\Omega(s)}{V(s)} = \frac{k_t}{(Ls+R)(Js+B)+k_t k_e}$$
$$= \frac{k_t}{LJ\,s^2 + (RJ+LB)\,s + (RB+k_t k_e)}$$

Disturbance path: $\dfrac{\Omega(s)}{T_L(s)} = -\dfrac{Ls+R}{(Ls+R)(Js+B)+k_t k_e}$

Position: $\dfrac{\Theta(s)}{V(s)} = \dfrac{1}{s}\,G_{\omega v}(s)$ (type-1, integrating)

Time constants ($B \approx 0$)

$\tau_e = \dfrac{L}{R}$ electrical

$\tau_m = \dfrac{R\,J}{k_t\,k_e}$ mechanical

$$G_{\omega v}(s) \approx \frac{1/k_e}{\tau_m \tau_e s^2 + \tau_m s + 1}$$
$$\overset{\tau_m \gg \tau_e}{\approx} \frac{1/k_e}{(\tau_m s+1)(\tau_e s+1)}$$

- Real poles if $\tau_m > 4\tau_e$; otherwise oscillatory.
- τ_m depends on **total** inertia: load inertia slows the motor.
- Typical: $\tau_e \approx$ 0.5–5 ms, $\tau_m \approx$ 2–20 ms.

Worked example: from parameters to poles

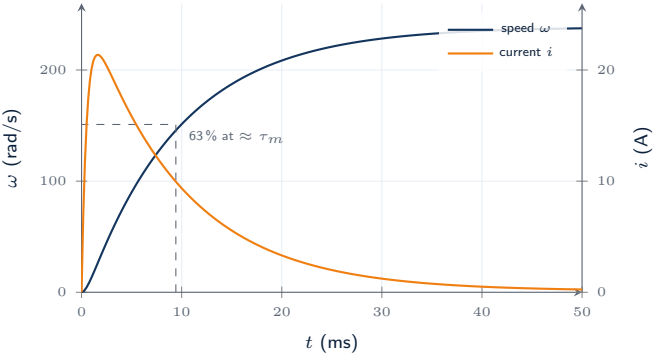
Parameter	Value
Supply v	48 V
R	1.0 Ω
L	0.5 mH
$k_t = k_e$	0.1 N m/A
J (rotor)	1.0×10^{-4} kg m^2
B	5×10^{-5} N m s/rad

1. $\tau_e = L/R = 0.5$ ms, $\quad \tau_m = RJ/(k_t k_e) = 10$ ms $(\tau_m/\tau_e = 20)$.
2. Denominator coefficients: $LJ = 5.0 \times 10^{-8}$, $\quad RJ + LB = 1.0003 \times 10^{-4}$, $RB + k_t k_e = 1.005 \times 10^{-2}$.
3. DC gain: $k_t/(RB + k_t k_e) = 9.95$ rad/(s V) $(\approx 1/k_e)$.
4. Poles: $s^2 + 2000.5\, s + 201\,000 = 0$
   $\Rightarrow s_1 = -106.1$/s $(\tau = 9.42$ ms $\approx \tau_m)$,
   $s_2 = -1894$/s $(\tau = 0.53$ ms $\approx \tau_e)$.
5. No-load speed at 48 V: $\omega_0 = 477.6$ rad/s $= 4561$ rpm.

Check

Approximate τ_m, τ_e are within 6 % of the exact pole time constants because $\tau_m \gg \tau_e$.

Step response of the example motor (24 V step, $T_L = 0$)



Exact solution (partial fractions):

$$\omega(t) = 238.8 + 14.2\,e^{-1894t} - 253.0\,e^{-106.1t}$$

$$i(t) = 0.12 - 26.8\,e^{-1894t} + 26.7\,e^{-106.1t}$$

- Current peaks at **21.4 A** after 1.6 ms – 180× the no-load current!
- Back-EMF builds up and pulls the current down: inherent damping.
- Real drives **limit the current**; with a current limit the speed ramps linearly ($\dot{\omega} = k_t I_{\max}/J$).

From datasheet to model – units and pitfalls

Datasheet item	Typical unit	Model value
Back-EMF const.	V/krpm	$\div\, 104.7 \rightarrow \mathrm{V\,s/rad}$
Torque const.	N m/A (rms!)	k_t
Terminal resistance	Ω line–line	$R_{\mathrm{ph}} = R_{LL}/2$ (Y)
Terminal inductance	mH line–line	$L_{\mathrm{ph}} = L_{LL}/2$ (Y)
Rotor inertia	$\mathrm{kg\,cm^2}$	$\times 10^{-4} \rightarrow \mathrm{kg\,m^2}$
Mech. time const.	ms	$\tau_m = RJ/(k_t k_e)$
Thermal resistance	K/W	R_{th} (Session 1.5)

Three-phase PMSM: watch the convention

With k_e in $\mathrm{V_{rms,LL}/(rad/s)}$ and k_t in $\mathrm{N\,m/A_{rms}}$ (phase current):

$$k_t = \sqrt{3}\, k_e$$

$k_t = k_e$ holds only in **consistent per-phase / dq units**.

- Example: $33\,\mathrm{V_{rms}/krpm} \Rightarrow k_e = 0.315$, $k_t = 0.546\,\mathrm{N\,m/A_{rms}}$.
- Hot winding: R rises $\approx 0.39\,\%/\mathrm{K}$ (copper): $+39\,\%$ at $\Delta T = 100\,\mathrm{K}$.
- Datasheet values have **±5–10 %** tolerance.

Session 1.3 – Summary / key take-aways

- Motor = RL circuit + back-EMF, coupled to an inertia with friction and load torque.
- Energy conservation gives $k_t = k_e$ in SI units.
- Transfer function $V \to \Omega$ is 2nd order with $\tau_e = L/R$ and $\tau_m = RJ/(k_t k_e)$.
- Friction: viscous (linear) + Coulomb/stiction (nonlinear); T_L is a disturbance.
- Without current control, a voltage step causes large inrush current.

Remember

$$v = Ri + L\dot{i} + k_e\omega$$

$$J\dot{\omega} = k_t i - B\omega - T_L$$

$$\frac{\Omega}{V} = \frac{k_t}{(Ls+R)(Js+B) + k_t k_e}$$

$$\tau_e = L/R, \quad \tau_m = RJ/(k_t k_e)$$

Session 4

1.4 Actuators and motion transmission

Motion transmission options

Mechanism	Conversion constant	Speed	Stiffness	Accuracy / backlash	Stroke
Ball screw	lead $p =$ 5–40 mm/rev	$\leq$ 1–2 m/s	high	μm, preloaded	$\lesssim$ 3–4 m (whip)
Toothed belt	pulley radius r	$\leq$ 5–10 m/s	low–medium	0.05–0.1 mm	long
Rack & pinion	pinion radius r	2–5 m/s	medium	backlash (dual pinion)	unlimited
Linear motor	direct, k_f in N/A	$>$5 m/s	servo-defined	nm–μm (encoder)	unlimited
Gearbox	ratio $N =$ 3–160	motor-limited	medium	1–10 arcmin	rotary
Direct-drive torque motor	1:1	low–medium	highest	no backlash	rotary

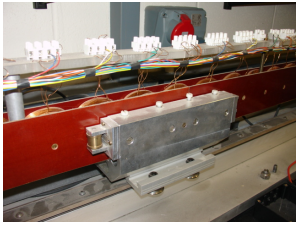
Geared / converted drive

+ small, fast motor; load inertia reduced by $1/N^2$
− backlash, compliance, friction, wear, noise

Direct drive

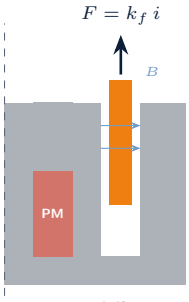
+ stiff, zero backlash, high bandwidth, low maintenance
− large, expensive motor; full load inertia and disturbances act on the motor

Linear electromechanical actuators: linear motor and voice coil



Zureks, CC BY-SA 4.0, via Wikimedia Commons

- **Linear motor** = PMSM "unrolled": forcer (coil) + magnet track.
- **Iron-core**: high force, cogging, attraction force $\approx$ 3–10$\times$ thrust.
- **Ironless** (U-channel): no cogging, lower force, very smooth.

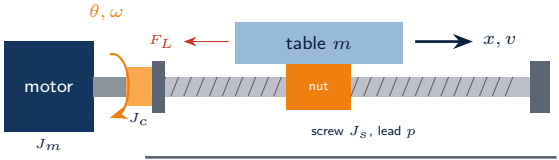


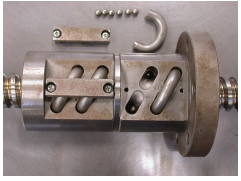
axisymmetric half-section

$$F = k_f\, i, \qquad e = k_f\, v$$
$$k_f = N\, B\, l \quad [\mathrm{N/A} = \mathrm{V\,s/m}]$$
$$m\,\ddot{x} = k_f\, i - b\,\dot{x} - F_L$$

- **Voice coil**: moving coil in a radial field – no commutation, no cogging, stroke mm to a few cm.
- Very high bandwidth: HDD head positioners, lens focus, fast tool servos, active vibration isolation.
- Same equations as the DC motor with $T \to F$, $\omega \to v$, $J \to m$.

Ball screw: kinematics and equivalent inertia





Glenn McKechnie (Graibeard), CC BY-SA 3.0, via Wikimedia Commons

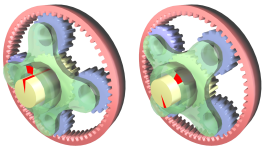
$$x = \frac{p}{2\pi}\,\theta, \qquad v = \frac{p}{2\pi}\,\omega$$

$$T = \frac{p}{2\pi\,\eta}\,F \qquad \text{(driving)}$$

$$J_{\text{eq}} = J_m + J_c + J_s + m\left(\frac{p}{2\pi}\right)^2$$

- Derivation: $\frac{1}{2}mv^2 = \frac{1}{2}\,m\left(\frac{p}{2\pi}\right)^2\omega^2$ (kinetic energy).
- Screw inertia: $J_s = \pi\rho L d^4/32$ – grows with d^4!
- $\eta \approx 0.9$ (ball screw) vs. 0.3–0.6 (sliding lead screw).
- Limits: critical (whip) speed, buckling, nut DN value, axial stiffness.

Gearboxes: planetary and harmonic (strain-wave) drives



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Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

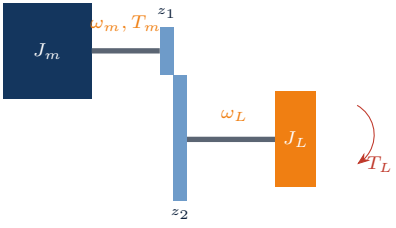
Planetary (ring fixed, sun in, carrier out): $N = 1 + \dfrac{Z_{\text{ring}}}{Z_{\text{sun}}}$

Harmonic (circular spline fixed, wave generator in, flexspline out): $N = \dfrac{Z_{\text{fs}}}{Z_{\text{cs}} - Z_{\text{fs}}}$ (reversed)

Example: $Z_{\text{fs}} = 200$, $Z_{\text{cs}} = 202 \Rightarrow N = 100$ in one stage.

	Ratio / stage	Efficiency	Backlash	Torsional stiffness	Typical use
Planetary	3–10	95–98 %	1–10 arcmin	high	servo axes, gantries
Harmonic drive	30–160	60–90 %	≈ 0	low, nonlinear	robot joints
Cycloidal	30–120	80–90 %	$<$ 1 arcmin	very high	heavy robot axes
Worm	5–100	40–90 %	medium	medium	self-locking drives

Reflected inertia and load torque through a gear



Kinetic energy must be preserved:

$$\tfrac{1}{2} J_L\, \omega_L^2 = \tfrac{1}{2} \underbrace{\frac{J_L}{N^2}}_{J'_L} \omega_m^2$$

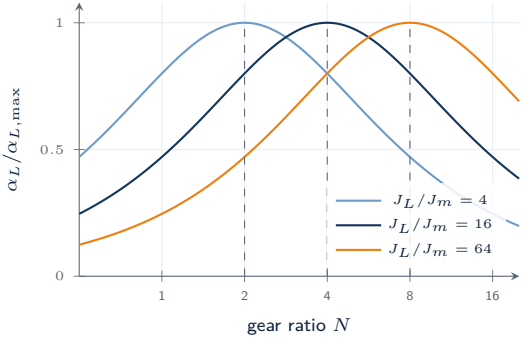
Power balance with efficiency η:

$$T'_L\, \omega_m\, \eta = T_L\, \omega_L \Rightarrow T'_L = \frac{T_L}{N\eta}$$

$$T_m = \left(J_m + J_g + \frac{J_L}{N^2}\right)\dot{\omega}_m + \frac{T_L}{N\eta}$$

Mechanism	Reflected inertia at motor
Gear ratio N	J/N^2
Ball screw, lead p	$m\,(p/2\pi)^2$
Belt / rack, radius r	$m\,r^2$ (+ pulleys/pinion)

Inertia ratio and the optimal gear ratio



Load acceleration for a given motor torque T_m (no friction):

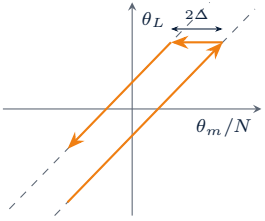
$$\alpha_L = \frac{N\,T_m}{J_m N^2 + J_L}$$

$d\alpha_L/dN = 0$ gives

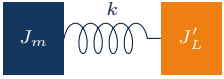
$$N_{\text{opt}} = \sqrt{J_L/J_m} \;\Leftrightarrow\; J_L/N^2 = J_m \text{ (inertia matching)}$$

- The maximum is flat: N within $\pm 30\,\%$ of N_{opt} costs $< 6\,\%$.
- In practice N is set by **speed**: $N \le \omega_{m,\max}/\omega_{L,\max}$.
- **Inertia ratio** J_L'/J_m: guideline $\le$ 3–10 for good servo tuning with compliant mechanics; direct drives tolerate more.
- Large ratios $\Rightarrow$ low mechanical resonance relative to bandwidth, sensitive tuning (Day 5).

Transmission non-idealities: efficiency, backlash, stiffness



backlash: dead zone of width 2Δ



$$f_r = \frac{1}{2\pi}\sqrt{\frac{k\,(J_m + J'_L)}{J_m\,J'_L}}$$

compliant gear/coupling/belt $\Rightarrow$ resonance (Session 2.1, Day 5)

- **Efficiency** η: losses $\propto$ torque; back-driving efficiency differs (worm/lead screw may self-lock).
- **Backlash** Δ: lost motion at reversal, limit cycles in position loops; reduce by preload, dual pinions, harmonic drives.
- **Stiffness** k: belts and harmonic drives are compliant; ball screws stiff axially but screw torsion/nut stiffness matter.
- Load-side encoder "sees" backlash and compliance – dual-loop control (Day 5).

Session 1.4 – Summary / key take-aways

- Transmissions trade motor size against stiffness, backlash and friction.
- Every load is reflected to the motor shaft by **energy equivalence**.
- Optimal (acceleration) gear ratio: $N_{\text{opt}} = \sqrt{J_L/J_m}$; in practice speed limits N.
- Keep the reflected inertia ratio moderate; compliant drives need careful tuning.
- Linear motors and voice coils obey the same equations with F, v, m.

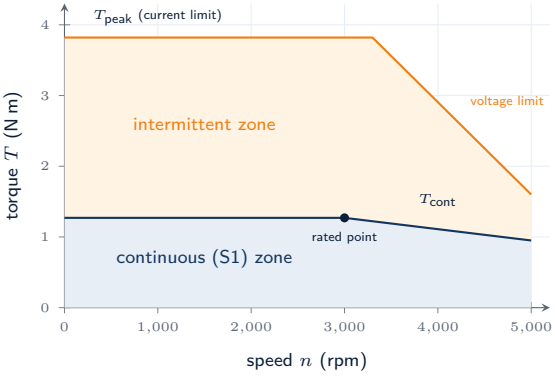
Remember

$$J'_L = J_L/N^2, \quad T'_L = T_L/(N\eta)$$
$$J'_{\text{screw}} = m\,(p/2\pi)^2, \quad J'_{\text{belt}} = m\,r^2$$
$$N_{\text{opt}} = \sqrt{J_L/J_m}$$
$$f_r = \tfrac{1}{2\pi}\sqrt{k(J_m + J'_L)/(J_m J'_L)}$$

Session 5

1.5 Characteristics and selection criteria

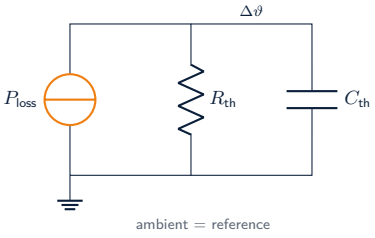
Servo motor torque–speed envelope



Generic 400 W AC servo class: $T_N = 1.27$ N m, $T_{\text{peak}} = 3.82$ N m, $n_N = 3000$ rpm.

- **Continuous torque** T_{cont}: limited by **winding temperature** at steady state (S1); decreases at speed (iron and friction losses).
- **Peak torque** T_{peak}: limited by drive current, magnet demagnetisation and saturation; allowed for short times only.
- **Voltage limit**: at high speed the back-EMF leaves no voltage reserve to force current $\Rightarrow$ T_{peak} drops. Depends on **DC bus voltage**.
- Curves are specified for a given ambient (40 °C), mounting flange and drive.

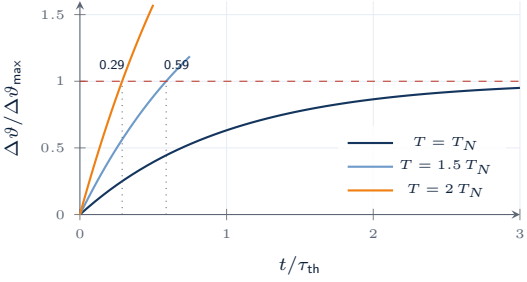
Thermal limits: thermal model and time constant



$$C_{\text{th}}\,\frac{d\Delta\vartheta}{dt} = P_{\text{loss}} - \frac{\Delta\vartheta}{R_{\text{th}}}$$

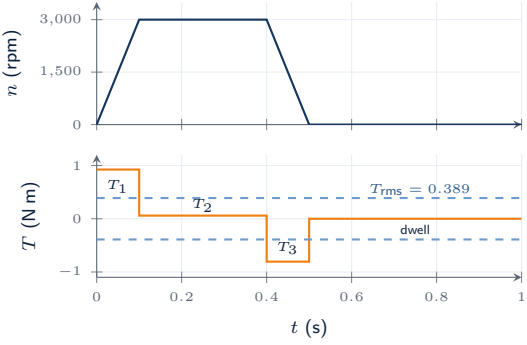
$$\Delta\vartheta(t) = P\,R_{\text{th}}\left(1 - e^{-t/\tau_{\text{th}}}\right)$$

$$\tau_{\text{th}} = R_{\text{th}}\,C_{\text{th}}$$



- Copper loss $\propto T^2$: overload $k = T/T_N$ reaches the limit after $t = -\tau_{\text{th}}\ln(1 - 1/k^2)$.
- τ_{th} (motor) $\approx$ 10–60 min, but the **winding** has seconds-scale time constants: peak torque only for $\approx$ 0.1–few s.
- Insulation class F/H (155/180 °C); magnets degrade if too hot. Drives use I^2t models and sensors (PTC/KTY/Pt1000).

Duty cycle and RMS torque



Copper losses $\propto i^2 \propto T^2$. If the cycle time $t_{\text{cyc}} \ll \tau_{\text{th}}$, the winding sees the **mean loss**:

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum_k T_k^2 \, t_k} \;\le\; T_{\text{cont}}(\bar{n})$$

- Trapezoidal profile: $T_1 = J_{\text{tot}}\alpha + T_f$, $T_2 = T_f$, $T_3 = -J_{\text{tot}}\alpha + T_f$, dwell: $T_4 = T_{\text{hold}}$.
- Include holding torque on **vertical axes** (gravity) – dwell is not free!
- Duty types per IEC 60034-1: S1 continuous, S2 short-time, S3 intermittent periodic, ...
- Also check **peak** $\max|T_k| \le T_{\text{peak}}(n)$ and $n_{\max}$.

Worked sizing example (1/2): load and move profile

Given: horizontal ball-screw axis

Table + workpiece $m = 50\,\text{kg}$
Screw: lead $p = 10\,\text{mm}$, $d = 20\,\text{mm}$, $L = 0.8\,\text{m}$, steel, $\eta = 0.9$
Coupling $J_c = 0.10 \times 10^{-4}\,\text{kg m}^2$
Guide friction $\mu = 0.01$; bearings/seal $T_b = 0.05\,\text{N m}$
Move 200 mm: $t_a = t_d = 0.1\,\text{s}$, total 0.5 s, dwell 0.5 s ($t_{\text{cyc}} = 1\,\text{s}$)

1. Kinematics

$$v_{\max} = \frac{s}{t_a + t_c} = \frac{0.2}{0.4} = 0.5\,\text{m/s}, \quad n = \frac{v}{p} = 50\,\text{rev/s} = 3000\,\text{rpm}$$
$$\omega_{\max} = 314.2\,\text{rad/s}, \quad \alpha = \omega_{\max}/t_a = 3142\,\text{rad/s}^2$$

2. Inertia at the motor shaft

$$J_s = \pi\rho L d^4/32 = \pi \cdot 7850 \cdot 0.8 \cdot 0.02^4/32 = 0.99 \times 10^{-4}\,\text{kg m}^2$$
$$J_t = m\,(p/2\pi)^2 = 50\,(0.01/2\pi)^2 = 1.27 \times 10^{-4}\,\text{kg m}^2$$
$$J_L = J_s + J_t + J_c = 2.35 \times 10^{-4}\,\text{kg m}^2$$

3. Friction torque

$$T_f = \frac{\mu m g\, p}{2\pi\eta} + T_b = \frac{0.01 \cdot 50 \cdot 9.81 \cdot 0.01}{2\pi \cdot 0.9} + 0.05 = 0.059\,\text{N m}$$

> Assumption: efficiency applied to friction only, not to inertia (conservative alternative: J_t/η).

Worked sizing example (2/2): torques, RMS and check

4. Candidate: 400 W servo, $J_m = 0.40 \times 10^{-4}$ kg m^2, $T_N = 1.27$, $T_{\text{peak}} = 3.82$ N m, $n_N = 3000$ rpm
Inertia ratio $J_L/J_m = 5.9$ ✓ $J_{\text{tot}} = 2.75 \times 10^{-4}$ kg m^2

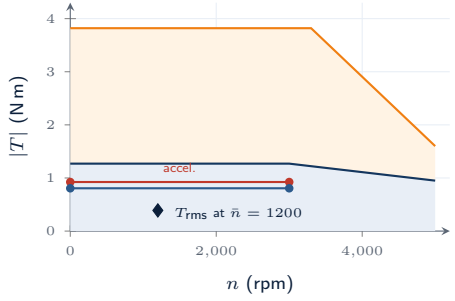
5. Segment torques
$J_{\text{tot}}\alpha = 2.75 \cdot 10^{-4} \cdot 3142 = 0.865$ N m
$T_1 = 0.865 + 0.059 = 0.924$ N m (0.1 s)
$T_2 = 0.059$ N m (0.3 s),
$T_3 = -0.865 + 0.059 = -0.806$ N m (0.1 s), $T_4 = 0$

6. RMS torque

$$T_{\text{rms}} = \sqrt{\frac{0.924^2 \cdot 0.1 + 0.059^2 \cdot 0.3 + 0.806^2 \cdot 0.1}{1.0}} =$$

0.389 N m

7. Checks: peak $0.924 < 3.82$ ✓; RMS $0.389 < 1.27$ (31 %) ✓; $n_{\max} = 3000 \le n_N$ ✓
Regenerated energy per stop: $\frac{1}{2} J_{\text{tot}} \omega^2 = 13.6$ J (bus capacitor / brake resistor).



A 200 W motor ($J_m = 0.18 \times 10^{-4}$, $T_N = 0.64$, $T_{\text{pk}} = 1.91$ N m) gives $T_{\text{rms}} = 0.358$, $T_{\text{pk}} = 0.854$ – thermally OK, but inertia ratio 13: **tuning/stiffness decides**.

Motor–drive matching and selection checklist

Voltage: the drive must supply back-EMF + drops at top speed

$$V_{\text{req}} \approx k_e\,\omega_{\max} + R\,I_{\text{pk}}\ (+\,\omega_e L\,I)\ \le\ V_{\text{avail}}(V_{\text{DC}})$$

Current: $I_{\text{pk}} = T_{\text{pk}}/k_t$, $I_{\text{rms}} = T_{\text{rms}}/k_t \le$ drive peak / continuous ratings.

Example (motor above, $k_e = 33\,\text{V}_{\text{rms,LL}}/\text{krpm}$, $k_t = 0.546\,\text{N m/A}$):

- EMF at 3000 rpm: $99\,\text{V}_{\text{rms,LL}}$ $\Rightarrow V_{\text{DC}} \gtrsim \sqrt{2}\cdot 99 \approx 140\,\text{V}$ + margin.
- 230 V drive ($V_{\text{DC}} \approx 325\,\text{V}$): ✓; 48 V drive: $n_{\max} \approx 1000\,\text{rpm}$ × ⇒ low-voltage winding needed.
- $I_{\text{pk}} = 0.924/0.546 = 1.7\,\text{A}_{\text{rms}}$, $I_{\text{rms}} = 0.71\,\text{A}_{\text{rms}}$.

Same motor frame, different winding: $k_t \uparrow \Leftrightarrow$ less current, but higher voltage per speed.

Selection checklist

1. Define move profile, loads, cycle, environment.
2. Reflect inertia/torques to the motor (gear, screw, belt).
3. Choose ratio: speed range, N_{opt}, resolution.
4. Peak torque at all speeds inside the **intermittent zone**.
5. T_{rms} at mean speed inside the **continuous zone** (with 20–30 % margin).
6. Inertia ratio acceptable for the mechanics' stiffness.
7. Drive: voltage, peak/continuous current, regen energy.
8. Feedback resolution, brake, IP class, standards (IEC 60034, IEC 61800).

Session 1.5 – Summary / key take-aways

- The torque–speed envelope has a **thermal** (continuous) and a **current/voltage** (intermittent) boundary.
- Heat accumulates with τ_{th} of minutes: judge with RMS torque over the cycle.
- Peak torque must be available at the speed where it is needed.
- Motor and drive are matched through k_t, k_e, bus voltage and current ratings.
- Size systematically: kinematics $\rightarrow$ inertias $\rightarrow$ torques $\rightarrow$ RMS/peak $\rightarrow$ drive.

Remember

$$T_{\text{rms}} = \sqrt{\frac{1}{t_{\text{cyc}}} \sum T_k^2 t_k}$$

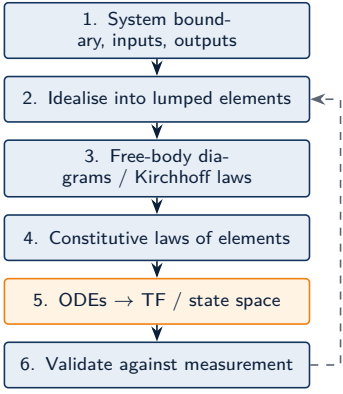
$$\Delta\vartheta = P R_{\text{th}} (1 - e^{-t/\tau_{\text{th}}})$$

$$I_{\text{pk}} = T_{\text{pk}}/k_t, \quad V \gtrsim k_e \omega_{\text{max}} + R I_{\text{pk}}$$

Session 6

2.1 Modelling of dynamic systems – fundamental physical models

Lumped-parameter modelling: workflow and building blocks

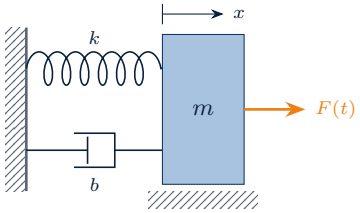


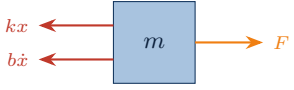
Domain	Kinetic store	Potential store	Dissipator
Translation	mass m	spring k	damper b
Rotation	inertia J	torsion spring k_θ	damper B
Electrical	inductor L	capacitor C	resistor R

- **Lumped**: wave propagation neglected – valid for frequencies well below the first ignored structural mode.
- Each independent energy store adds one state ($\Rightarrow$ system order).
- **Transformers** (gears, screws) and **gyrators** (motors: $T = ki$, $e = k\omega$) connect domains without storing energy.
- Choose the **simplest model that answers the question** (Day 2: linearisation, identification).

[Karnopp, Margolis & Rosenberg, 2012; Ogata, 2010]

Mass–spring–damper: free-body diagram to transfer function





free-body diagram

Newton: $\sum F = m\ddot{x}$ (positive x to the right):

$$m\,\ddot{x} + b\,\dot{x} + k\,x = F(t)$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{1}{k}\,\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Characteristic parameters

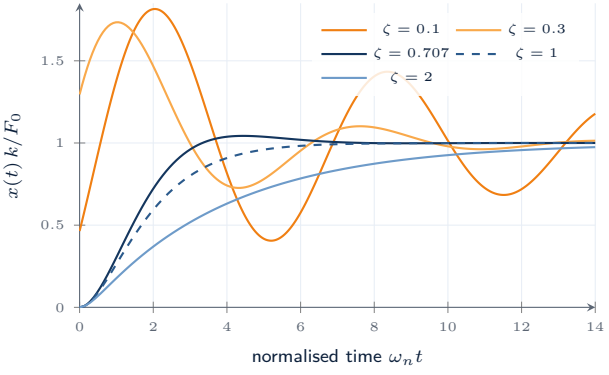
$\omega_n = \sqrt{k/m}$ natural frequency

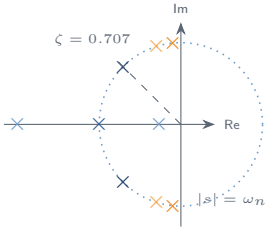
$\zeta = \dfrac{b}{2\sqrt{km}}$ damping ratio

$\omega_d = \omega_n\sqrt{1-\zeta^2}$ damped frequency

Example: $m = 2\,\text{kg}$, $k = 2 \times 10^5\,\text{N/m}$, $b = 120\,\text{N s/m} \Rightarrow$ $\omega_n = 316\,\text{rad/s}$ (50.3 Hz), $\zeta = 0.095$.

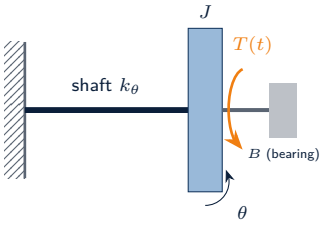
Second-order step responses and pole locations





- Poles $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$.
- Overshoot $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$: 73 %, 37 %, 4.3 %.
- Lightly damped mechanics ($\zeta < 0.1$) is the norm in servo structures.

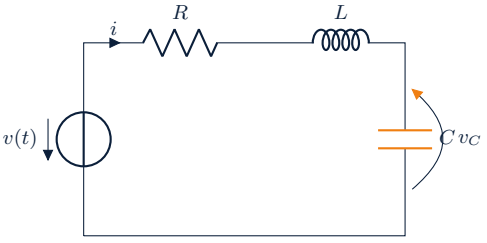
Rotational mechanical systems



$$J\,\ddot{\theta} + B\,\dot{\theta} + k_\theta\,\theta = T(t)$$
$$\frac{\Theta(s)}{T(s)} = \frac{1}{Js^2 + Bs + k_\theta}$$

- Euler's law: $\sum T = J\ddot{\theta}$ about a fixed axis.
- Torsional stiffness of a solid shaft: $k_\theta = \dfrac{G\,I_p}{l} = \dfrac{G\,\pi d^4}{32\,l}$.
- Example: steel ($G = 80\,\text{GPa}$), $d = 10\,\text{mm}$, $l = 100\,\text{mm}$ $\Rightarrow k_\theta = 785\,\text{N m/rad}$.
- Disc inertia $J = \frac{1}{2}mr^2$, hollow cylinder $J = \frac{1}{2}m(r_o^2 + r_i^2)$.
- Couplings, gear meshes and belts are springs too: series stiffness $1/k = \sum 1/k_i$ (softest element dominates).
- Servo motor without load: $k_\theta = 0 \Rightarrow \Theta/T = 1/(s(Js + B))$ – a double integrator for $B \to 0$.

Electrical RLC network



$$L\,\ddot{q} + R\,\dot{q} + \frac{1}{C}\,q = v(t), \qquad i = \dot{q}$$

$$\frac{V_C(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}$$

- KVL around the loop: $v = v_R + v_L + v_C$; constitutive laws $v_R = Ri$, $v_L = L\,di/dt$, $i = C\,dv_C/dt$.
- Same structure as the mass–spring–damper: $\omega_n = \dfrac{1}{\sqrt{LC}}, \quad \zeta = \dfrac{R}{2}\sqrt{\dfrac{C}{L}}.$
- In drives: DC-link LC input filter, sine/dv/dt output filters, cable resonances.
- Example: $L = 1\,\text{mH}$, $C = 470\,\mu\text{F}$, $R = 0.1\,\Omega$ $\Rightarrow f_n = 232\,\text{Hz}$, $\zeta = 0.034$ – a lightly damped filter.
- Motor winding alone ($C \to \infty$): first-order $1/(Ls + R)$, $\tau_e = L/R$.

Analogies and energy-based modelling

Translational	Rotational	Force–voltage	Force–current
force F	torque T	voltage v	current i
velocity $\dot{x}$	speed ω	current i	voltage v
mass m	inertia J	inductance L	capacitance C
damper b	damper B	resistance R	conductance $1/R$
spring k	torsion k_θ	elastance $1/C$	reciprocal ind. $1/L$
$\frac{1}{2}m\dot{x}^2$	$\frac{1}{2}J\omega^2$	$\frac{1}{2}Li^2$	$\frac{1}{2}Cv^2$
$\frac{1}{2}kx^2$	$\frac{1}{2}k_\theta\theta^2$	$\frac{1}{2}q^2/C$	$\frac{1}{2}\lambda^2/L$

Product of **effort** and **flow** is always **power**: Fv, $T\omega$, vi.

Lagrange's equations

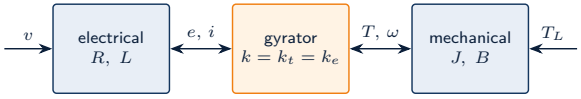
$\mathcal{L} = T_{\text{kin}} - V_{\text{pot}}$, $D =$ Rayleigh dissipation

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \frac{\partial \mathcal{L}}{\partial q_j} + \frac{\partial D}{\partial \dot{q}_j} = Q_j$$

MSD: $\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$, $D = \frac{1}{2}b\dot{x}^2$, $Q = F$
$\Rightarrow m\ddot{x} + b\dot{x} + kx = F$ ✓

- Energy view gives the model **without free-body diagrams** – useful for multi-body and coupled systems.
- Same Lagrangian approach with $q =$ charge yields the RLC equation.

Electromechanical coupling: the motor as a gyrator



$T = k\,i, \quad e = k\,\omega$
(effort $\leftrightarrow$ flow swapped)

State $\boldsymbol{x} = [i \ \ \omega]^\mathsf{T}$:

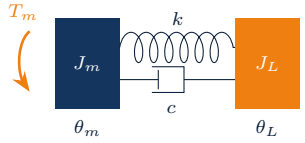
$$\frac{d}{dt}\begin{bmatrix} i \\ \omega \end{bmatrix} = \begin{bmatrix} -R/L & -k/L \\ k/J & -B/J \end{bmatrix}\begin{bmatrix} i \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L & 0 \\ 0 & -1/J \end{bmatrix}\begin{bmatrix} v \\ T_L \end{bmatrix}$$

Energy balance of the coupled system:

$$v\,i = \underbrace{R i^2 + B\omega^2}_{\text{dissipated}} + \frac{d}{dt}\underbrace{\left(\tfrac{1}{2}Li^2 + \tfrac{1}{2}J\omega^2\right)}_{\text{stored}} + T_L\,\omega$$

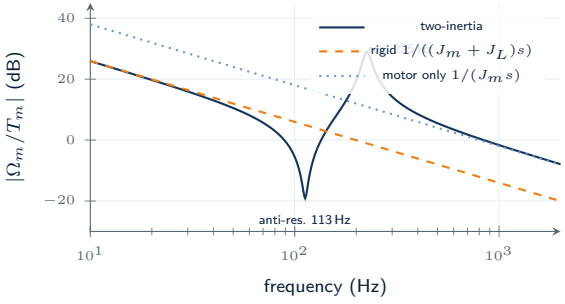
- The coupling terms ($\pm k$) form a **skew-symmetric** (power-conserving) interconnection: they exchange energy, never create or dissipate it.
- Seen from the terminals, inertia J looks like a **capacitance** J/k^2 and damping B like a conductance B/k^2.
- Two energy stores $\Rightarrow$ 2nd-order system, as found in Session 1.3.
- State-space form in detail: Day 2 (session 2.2).

Preview: two-inertia system (motor – compliant shaft – load)



$$J_m\ddot{\theta}_m = T_m - k(\theta_m - \theta_L) - c(\dot{\theta}_m - \dot{\theta}_L)$$
$$J_L\ddot{\theta}_L = k(\theta_m - \theta_L) + c(\dot{\theta}_m - \dot{\theta}_L) - T_L$$

$$\frac{\Omega_m}{T_m} = \frac{J_L s^2 + cs + k}{s\left[J_m J_L s^2 + c(J_m + J_L)s + k(J_m + J_L)\right]}$$



$J_m = 2 \times 10^{-4}$, $J_L = 6 \times 10^{-4}$ kg m^2, $k = 300$ N m/rad, $c = 0.02$ N m s/rad

- Anti-resonance $\omega_{ar} = \sqrt{k/J_L}$; resonance $\omega_r = \sqrt{k(J_m + J_L)/(J_m J_L)}$.
- Limits achievable servo bandwidth; motivates notch filters (Day 3, Day 5).

Session 2.1 (part 1) – Summary / key take-aways

- Lumped models: stores of kinetic and potential energy plus dissipators; each store adds a state.
- Mass–spring–damper, torsional system and RLC circuit share one 2nd-order form with ω_n, ζ.
- Analogies (force–voltage, force–current) let us reuse intuition and tools across domains.
- The motor is a gyrator: $T = ki$, $e = k\omega$; it couples the RL circuit to the inertia.
- Compliance between motor and load creates resonance and anti-resonance.

Remember

$$m\ddot{x} + b\dot{x} + kx = F$$

$$\omega_n = \sqrt{k/m}, \quad \zeta = b/(2\sqrt{km})$$

$$\omega_{ar} = \sqrt{k/J_L}$$

$$\omega_r = \sqrt{k(J_m + J_L)/(J_m J_L)}$$

Day 1 summary

Machines

Lorentz/Faraday $\Rightarrow$ $k_t = k_e$; reluctance $\frac{1}{2} i^2 dL/d\theta$.
DC, BLDC (six-step), PMSM (sinusoidal), induction (slip), stepper (open loop).

Models

$v = Ri + L\dot{i} + k_e\omega, \quad J\dot{\omega} = k_t i - B\omega - T_L$
$\tau_e = L/R$, $\tau_m = RJ/(k_t k_e)$; friction is nonlinear.

Transmissions and sizing

J/N^2, $m(p/2\pi)^2$, mr^2; $N_{\text{opt}} = \sqrt{J_L/J_m}$.
Peak in intermittent zone, T_{rms} in continuous zone; match V_{DC}, k_e, currents.

Dynamic modelling

Lumped elements, FBD, analogies, Lagrange; 2nd-order prototype ω_n, ζ; two-inertia resonance.

One sentence

A servo axis is an **energy converter** (motor) plus a **mechanical network** (transmission, load) – both described by the same lumped, energy-based equations.

Preview of Day 2: System dynamics and control

Code	Session	Time
2.1	Modelling (cont.): nonlinear effects and linearisation	0.5 h
2.2	Transfer functions and state space	1.0 h
2.3	System identification and frequency response	1.0 h
2.4	PID and basic control structures	1.5 h
2.5	Stability and performance analysis	1.0 h

Bring along from today:
the DC-motor model and its transfer function, the two time constants, the friction nonlinearity, and the 2nd-order prototype with ω_n and ζ – they are the plants we will linearise, identify and control tomorrow.

Figure credits

Slide	File (Wikimedia Commons)	Author	License
Brushed DC motor	Motor disassembeled.JPG	Jfer91	CC BY-SA 3.0
BLDC / PMSM	Floppy drive spindle motor open.jpg	Sebastian Koppehel	CC BY 3.0
Induction motor	Wirnik by Zureks.jpg	Zureks	CC BY-SA 3.0
Stepper motor	Positioner Stepper Motor parts.jpg	tony_duell	CC BY 2.0
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Ball screw	BallScrewNut-showing-balls-recirculating-tubes.jpg	Glenn McKechnie (Graibeard)	CC BY-SA 3.0
Gearboxes	Epicyclic gear ratios.png	Wapcaplet	CC BY-SA 3.0
Gearboxes	Harmonic Drive AG strain wave gear set.jpg	Pieceofmetalwork	CC BY-SA 4.0

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